Group I (revised file): check, and the running-coupling cancellation between Row II and Row V

The revised `Group_I.tex` was compared line by line with the version checked in `Group_I_check.pdf` (mistakes G1–G6 there). The changed equations were checked again numerically against first principles (`check_v2.py`). Equation numbers refer to the compiled revised file.

1 Summary

Fixed since the last check (verified).

- **G1 (Row III).** The fourth line of (1.10) now has $-\frac{ig^4 f^{abd}}{4\pi^4}$, and N_2 has the overall minus.
  - Eq. (1.10) now agrees with first principles at every p^+ (rel. 3×10^{-14}).
  - The new N_2 agrees with the principal-value integral of the new fourth line (rel. 3×10^{-14}).
  - The green terms ($\frac{1}{\Lambda}$, $\frac{1}{\vee-k^+}$, $\frac{1}{k^+-\Lambda}$) of N_1, N_2 and N_{3a} now cancel exactly (sum/term $= 6\times10^{-16}$). They can be removed.
- **G2 (Row II, part 2).** Eq. (1.3) now has $-k^i\log\frac{\vee}{k^+}$, the colour factor $f^{hbc}U^{ah}(w)\{\cdot\} - f^{abc}U^{bd}U^{ce}\{\cdot\}$ and $\int_{x'}$. The bracket equals $\mathbb{A}_2^{(3)}$ of `Calculation_main.tex` (rel. 2×10^{-15}). The colour factor of (1.8) is fixed too.
- **G6 (part).** $S^{ad}\to U^{ad}$ in the identity.
- **Row V, 2b finite, final form**, Eqs. (1.31) and (1.35). This is the corrected form: the B_2 pieces have $U(x)$, the B_4 endpoint has $e^{-ik\cdot(w'-z)}$, and the counterterm has $[2L-\frac{11}{6}]$ and $U(z)$. It agrees with the exact p^+ integral up to $O(\Lambda/k^+)$ (rel. 5.9×10^{-3} at $\Lambda/k^+=10^{-3}$, 5.2×10^{-6} at 10^{-5}).

Errors still in the file.

R1 New: the "Simplifying" equation (1.30) of $d^3N_{2b\,\text{finite}}$ (before the plus prescription). The two q^+ blocks of B_2 have the colour factor $[N_cU(y)-N_cU(z)]$. It must be $[N_cU(y)-N_cU(x)]$: the B_2 term belongs to 2b, whose colour is $N_c[U(y)-U(x)]$; only the counterterm has $U(z)$.

- As written, Eq. (1.30) differs from Eq. (1.29) by 0.10–1.13 (relative).
- With $U(x)$ it equals it exactly (3×10^{-13}).
- Your final equations (1.31), (1.35) already have $U(x)$; only this intermediate step is wrong.

R2 G3 is still open: the commutator $[\mathbb{N}_2,\mathbb{A}^{(1)}]$ in Row II part 2a, Eq. (1.5). In the $U^{ah}(w)f^{hbc}$ term, the $\log\frac{\vee}{\Lambda}$ must multiply only $\mathfrak{F}^i_\beta$, the first line of $\mathfrak{F}^i$:

$$\Big[\log\frac{\vee}{\Lambda}\,\mathfrak{F}^i\,\mathfrak{F}^i_\beta - \log\frac{\vee}{k^+}\,\mathfrak{F}^i\Big]U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\} - \Big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}\Big]\mathfrak{F}^i f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\},$$

with $\mathfrak{F}^i_\beta = \frac{(x-w)^i}{(x-w)^2}\int_z\frac{(w-z)\cdot(y-z)}{(w-z)^2(y-z)^2} - \frac{(y-w)^i}{(y-w)^2}\int_z\frac{(w-z)\cdot(x-z)}{(w-z)^2(x-z)^2}$. This changes only a three-ρ $\log\frac{\vee}{\Lambda}$ term; it is needed for the three-ρ cancellation (fix F4). See `Group_I_check.pdf`, Sec. G3.

R3 G4 is still open: principal values in Row III. In N_1, N_2, N_{3a}, every logarithm whose argument contains $k^+(x-w)^2-c\,(x-z)^2$ must be $\log|\cdot|$. When the pole $p^+_*=k^+(x-w)^2/(x-z)^2$ lies inside the p^+ range, the argument is negative.

R4 G5: limits that are not exact (only relevant if you keep Λ and $\vee$ exact).

- Row II, part 1 bracket (1.1), (1.2), and R^i (1.9) are the $\Lambda\to0$ forms. The exact ones are in `Group_I_corrected.pdf` (G5a, G5b).
- Row V part 1 (1.19), 2a (1.26) and the $[\frac{k^+}{q^+}]_+$ term of 2b finite, (1.31), use $\int_0^{k^+}$, which is exact only up to $O(\Lambda/k^+)$.

- Eqs. (1.28) and (1.34) (2b UV) drop $\frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}$ but keep $2\log(1-\frac{\Lambda}{k^+})$, which is of the same order.
- The same applies to the counterterm bracket $[-\frac{11}{6} + 2L]$ of 2b finite, (1.31), (1.35), where $L = \log\frac{k^+-\Lambda}{\Lambda}$ keeps Λ.

Either drop all $O(\Lambda/k^+)$ terms (then $L \to \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}$), or keep all of them.

R5 Notation (G6, still open):

- $\log[1-\frac{k^+}{\vee}]^2$ in (1.1), (1.2) means $\log^2(1-\frac{k^+}{\vee})$;
- $\int_{x'}$ is still missing in (1.8);
- the phase $e^{-ik\cdot(z-x)p^+/k^+}$ stands in front of $\int dp^+$ and $\int_{x,z}$ in Row V; it belongs inside;
- $\int_{x,z,y,x',y'w'}$ needs a comma;
- typos "Rewwriting", "crossection".

Everything else is as in `Group_I_check.pdf`: Row II part 1, the 2a/2b split, Row IV and Row V (part 1, part 2 up to 2b) are correct.

2 Running coupling: the cancellation between Row II and Row V

The β_0 terms of Group I are in two places:

- Row II, part 1: the $\frac{11}{3}$ in the bracket that multiplies the pole;
- Row V, part 2b (UV term): the $-\frac{11}{6}$.

Both multiply the LO structure, with the emitting charge and the gluon at different (dummy) points. Write $\mathcal{K}(v) \equiv \frac{v}{v^2}$ and $E(r) \equiv \frac{2}{\epsilon} + 2\gamma_E + \log\frac{r^2\mu^2_{\overline{MS}}}{4}$.

2.1 The two terms, per ordering

Row II, part 1 (your Eq. (1.1); the bracket there is $-E(x-w)$):

$$\left.\frac{d^3N_{\rm II}}{d^3k}\right|_{\beta_0} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\big[-E(x-w)\big]\,\frac{11}{3}$$
$$\times\,\big[U(x')-U(w')\big]^{ab'}\rho^{b'}(x')\big[U(w)-U(x)\big]^{ab}\rho^b(x). \tag{2.1}$$

Row V, 2b UV (your Eq. (1.27)):

$$\left.\frac{d^3N_{\rm V}}{d^3k}\right|_{\beta_0} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\Big(-\frac{11}{6}\Big)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\,E(y-z)\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)$$
$$\times\,\big[U(x')-U(w')\big]^{ab'}\big[U(y)-U(z)\big]^{af}\rho^{b'}(x')\rho^f(y). \tag{2.2}$$

2.2 The cancellation

In Eq. (2.2), rename the integration variables $y \to x$ (charge) and $z \to w$ (gluon), and the colour index $f \to b$. Then:

- the phase becomes $e^{-ik\cdot(w'-w)}$;
- the kernel becomes $\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)$, and $E(y-z) \to E(x-w)$;
- the colour factor becomes $[U(x')-U(w')]^{ab'}\rho^{b'}(x')[U(x)-U(w)]^{ab}\rho^b(x) = -[U(x')-U(w')]^{ab'}\rho^{b'}(x')[U(w)-U(x)]^{ab}\rho^b(x)$;
- the prefactor is $\frac{g^4N_c}{16\pi^4}\big(-\frac{11}{6}\big) = -\frac{g^4N_c}{32\pi^4}\frac{11}{3}$.

So, with the colour factor of Eq. (2.1),

$$\left.\frac{d^3N_{\rm V}}{d^3k}\right|_{\beta_0} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,E(x-w)\,\frac{11}{3}\,\big[U(x')-U(w')\big]\rho(x')\big[U(w)-U(x)\big]\rho(x)$$
$$= -\left.\frac{d^3N_{\rm II}}{d^3k}\right|_{\beta_0}. \tag{2.3}$$

$$\boxed{\left.\frac{d^3N_{\rm II}}{d^3k}\right|_{\beta_0} + \left.\frac{d^3N_{\rm V}}{d^3k}\right|_{\beta_0} = 0 \qquad \text{per ordering, and for the c.c.}} \tag{2.4}$$

- The cancellation is complete: the pole $\frac{2}{\epsilon}$, the $2\gamma_E$, the μ dependence, and the finite logarithm $\log(x-w)^2$ all cancel, since after renaming the distance is the same.
- Numerically, with random adjoint Wilson lines and charges, the sum over the 11/3 term is 2×10^{-16}.
- It is exact in Λ. The non-logarithmic parts of the exact brackets (G5a, G5d), $\left(\frac{11}{3}-8\delta+2\delta^2-\frac{4}{3}\delta^3\right)$ for Row II and $2\times\left(-\frac{11}{6}+4\delta-\delta^2+\frac{2}{3}\delta^3\right)$ for Row V 2b, with $\delta=\frac{\Lambda}{k^+}$, add up to 0 (`sympy`).

So this is the cancellation to carry out: **the $\frac{11}{3}$ term of Row II part 1 against the $-\frac{11}{6}$ term of Row V 2b (UV), after $y\to x$, $z\to w$**, ordering by ordering.

2.3 What remains in the two brackets

After this cancellation, the poles of Row II and Row V 2b multiply only rapidity logarithms. Per ordering, the coefficient of $\frac{\alpha_s N_c}{4\pi}E\times$LO is:

- Row II: $-2\log\frac{\vee}{\Lambda}+2\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})$;
- Row V 2b: $4\log\frac{\vee}{\Lambda}-4\log\frac{\vee}{k^+}+4\log(1-\frac{\Lambda}{k^+})$.

These are not running coupling.

- Their $\log\frac{\vee}{\Lambda}$ parts belong to the evolution (JIMWLK/BFKL) terms.
- Their poles combine with the soft endpoint of Group II, Row IV. With the c.c. the $\log\frac{\vee}{\Lambda}$ poles cancel with weights +2 (Row IV), +2 (Row II), −4 (Row V 2b) (`RowIV_FF_userform.pdf`, Sec. 4).

2.4 Consequences for the running coupling

- **No running-coupling logarithm survives in Group I.** The $\beta_0\log(\text{distance}^2\mu^2)$ terms of Row II and Row V cancel identically. This supersedes the "triumvirate" structure in Sec. 8.4 of the paper draft (`sec8_rc_dglap.tex`), which assumed a different net weight of Row V 2b.
- **Charge renormalization.**
  - The LO cross section is $\propto g^2$. With $\alpha_0=\alpha_s(\mu)\left[1-\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\right]$ ($\beta_0=\frac{11}{3}N_c$), its pole is not cancelled inside Group I.
  - It combines with the collinear pole of Group II Row IV into the full P_{gg}: it is the $\frac{\beta_0}{2}\delta(1-\xi)=\frac{11N_c}{6}\delta(1-\xi)$ of the fragmentation counterterm (`RowIV_FF_userform.pdf`).
- **Scale of the coupling.** The μ dependence of $\alpha_s(\mu)$ in LO, $-\frac{\alpha_s}{2\pi}\frac{\beta_0}{2}\times$LO per $\log\mu^2$, is compensated by the $\delta(1-\xi)$ part of the DGLAP evolution of the fragmentation function. The coupling of LO therefore runs with the fragmentation scale; the natural choice is $\mu\simeq\frac{2e^{-\gamma_E}}{|y'-y|}\sim|\mathbf{k}|$.
- **Finite constant.** Row II also contains $-\frac{67}{9}+\frac{\pi^2}{3}$, which is not multiplied by the pole. Per ordering this is $+\frac{\alpha_s}{2\pi}K\times$LO, with $K=N_c\left(\frac{67}{18}-\frac{\pi^2}{6}\right)$; with the c.c. it is $\frac{\alpha_s}{\pi}K\times$LO.
  - It is the Catani–Marchesini–Webber constant: it can be absorbed by using $\alpha_s^{\rm CMW}=\alpha_s(1+\frac{K\alpha_s}{2\pi})$.
  - Row V 2b (UV) has no such constant in this scheme; whether its finite part supplies one was not determined here.
  - This is not a logarithm, so it does not affect the scale of the coupling.

3 Checks

Script `check_v2.py` → `check_v2.json` (it reuses `fp1.py`, `user1.py` and `../rowV/part2_user_check.py`).

check	result
Row II part 2: new bracket vs. $\mathbb{A}_2^{(3)}$ of `Calculation_main.tex`	rel. 2×10^{-15}
Row III: new Eq. (1.10) vs. first principles (E3), symmetric and general U, 4 values of p^+	rel. 3×10^{-14}
Row III: new N_2 vs. principal-value integral	rel. 3×10^{-14}
Row III: sum of the green terms / one green term	6×10^{-16}
Row V 2b finite, "Simplifying", as written ($U(z)$ on B_2)	rel. 0.10–1.13
the same with $U(x)$ on B_2	rel. 3×10^{-13}
Row V 2b finite, final form, as written	rel. 5.9×10^{-3} ($\Lambda/k^+=10^{-3}$), 5.2×10^{-6} (10^{-5})
β_0: Row II (11/3) + Row V 2b UV (−11/6), random U, ρ	2×10^{-16}
β_0 and powers of Λ/k^+, exact brackets (`sympy`)	0